

GRADED SOLUTIONS:

```
1.  import psycopg2
    import sys
    import os
    f = open("date.txt", "r")
    database = sys.argv[1]
    user = os.environ.get('PGUSER')
    password = os.environ.get('PGPASSWORD')
    host = os.environ.get('PGHOST')
    port = os.environ.get('PGPORT')
    con = psycopg2.connect(
        dbname=database,
        host=host,
        user=user,
        password=password,
        port=port
    )
    cur = con.cursor()
    details = []
    for date in f.readlines():
        cur.execute(f"select referees.name from referees, match_referees, matches
        where matches.match_date='{date}' and
        match_referees.referee=referees.referee_id and
        match_referees.match_num=matches.match_num")
        details.append(cur.fetchone())
    def nameformat(name):
        a = name.split(" ")
        lname = a[-1]
        editedname = lname
        a.pop()
        for i in a:
            editedname+=f" {i[0]}."
        return editedname
    for name in details:
        formattedname = nameformat(name[0])
        print(formattedname)
    cur.close()
    con.close()
```

```
2. import psycopg2
import sys
import os
import math
f = open("parameter.txt", "r")
database = sys.argv[1]
user = os.environ.get('PGUSER')
password = os.environ.get('PGPASSWORD')
host = os.environ.get('PGHOST')
port = os.environ.get('PGPORT')
con = psycopg2.connect(
    dbname=database,
    host=host,
    user=user,
    password=password,
    port=port
)
cur = con.cursor()
details = []
for team in f.readlines():
    cur.execute(f"select sum(m.host_team_score) from teams t, matches m where
    t.team_id=m.host_team_id and t.name like '{team}%' and
    m.host_team_score>m.guest_team_score")
    details.append(cur.fetchone())
S = details[0][0]
X = S * 10
X_deg = X * math.pi / 180
print(round(math.cos(X_deg),2))
```

PRACTICE SOLUTIONS:

```
1. import sys
import os
import psycopg2
# Prime check
def is_prime(num):
    if num < 2:
        return False
    for i in range(2, num):
        if num % i == 0:
            return False
    return True
try:
    connection = psycopg2.connect(
        database=sys.argv[1],
        user=os.environ.get('PGUSER'),
        password=os.environ.get('PGPASSWORD'),
        host=os.environ.get('PGHOST'),
        port=os.environ.get('PGPORT')
    )
    cursor = connection.cursor()
    query = """
        SELECT p.name, t.name, p.jersey_no
        FROM players p
        JOIN teams t ON p.team_id = t.team_id
        ORDER BY p.name DESC, t.name DESC
    """
    cursor.execute(query)
    result = cursor.fetchall()

    # Filter by prime jersey numbers in Python
    for player_name, team_name, jersey_no in result:
        if is_prime(jersey_no):
            print(f"{player_name}, {team_name}")
    cursor.close()
except (Exception, psycopg2.DatabaseError) as error:
    print(error)
finally:
    if connection:
        connection.close()
```

```

2. import sys
import os
import psycopg2
# Encoding function
def encode_team_id(team_id):
    encoded = ""
    for ch in team_id:
        if ch.isalpha(): # Encode alphabets
            pos = ord(ch.upper()) - ord('A') + 1
            new_pos = (pos + 7) % 26
            if new_pos == 0:
                new_pos = 26
            new_char = chr(ord('A') + new_pos - 1)
            # preserve case
            if ch.islower():
                new_char = new_char.lower()
            encoded += new_char
        elif ch.isdigit(): # Encode digits
            new_digit = (int(ch) + 7) % 10
            encoded += str(new_digit)
        else:
            encoded += ch
    return encoded
try:
    connection = psycopg2.connect(
        database=sys.argv[1],
        user=os.environ.get('PGUSER'),
        password=os.environ.get('PGPASSWORD'),
        host=os.environ.get('PGHOST'),
        port=os.environ.get('PGPORT')
    )
    cursor = connection.cursor()
    # Query to get team ids where home and away jersey differ
    query = """
        SELECT team_id
        FROM teams
        WHERE jersey_home_color <> jersey_away_color
        ORDER BY team_id ASC;
    """
    cursor.execute(query)
    result = cursor.fetchall()

```

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    for row in result:
        team_id = row[o]
        print(encode_team_id(team_id))
    cursor.close()
except (Exception, psycopg2.DatabaseError) as error:
    print(error)
finally:
    if connection:
        connection.close()
```

```
3. import sys
import os
import psycopg2

file = open("player.txt", "r")
x = file.read()

try:
    connection = psycopg2.connect(
        database = sys.argv[1],
        user = os.environ.get('PGUSER'),
        password = os.environ.get('PGPASSWORD'),
        host = os.environ.get('PGHOST'),
        port = os.environ.get('PGPORT'))
    cursor = connection.cursor()
    query = "select jersey_no from players where name='%s'"%x
    cursor.execute(query)
    result = cursor.fetchall()
    for i in result:
        print(i[0])
    cursor.close()
except(Exception, psycopg2.DatabaseError) as error:
    print(error)
finally:
    connection.close()
```

```
4. import sys
import os
import psycopg2

file=open("name.txt","r")
x=file.read()

try:
    connection=psycopg2.connect(
        database=sys.argv[1],
        user=os.environ.get('PGUSER'),
        password=os.environ.get('PGPASSWORD'),
        host=os.environ.get('PGHOST'),
        port=os.environ.get('PGPORT'))
    cursor=connection.cursor()
    query="select roll_no from students where student_fname='%s'"%x
    cursor.execute(query)
    result=cursor.fetchall()
    for i in result:
        print(i[0])
    cursor.close()
except(Exception,psycopg2.DatabaseError) as error:
    print(error)
finally:
    connection.close()
```

```
5. import sys,os,psycopg2
file=open("team.txt","r")
x=file.read()

try:
    connection=psycopg2.connect(
        database=sys.argv[1],
        user=os.environ.get('PGUSER'),
        password=os.environ.get('PGPASSWORD'),
        host=os.environ.get('PGHOST'),
        port=os.environ.get('PGPORT'))
    cursor=connection.cursor()
    query="select playground from teams where team_id='%s'"%x
    cursor.execute(query)
    result=cursor.fetchall()
    for i in result:
        print(i[0])
    cursor.close()
except(Exception,psycopg2.DatabaseError) as error:
    print(error)
finally:
    connection.close()
```

```

6. import sys
import os
import psycopg2

file = open("name.txt", "r")
x = file.read()

def odd_even(num):
    if num%2==0:
        return "Even"
    else :
        return "Odd"

try:
    connection = psycopg2.connect(
        database = sys.argv[1],
        user = os.environ.get('PGUSER'),
        password = os.environ.get('PGPASSWORD'),
        host = os.environ.get('PGHOST'),
        port = os.environ.get('PGPORT'))
    cursor = connection.cursor()
    query = "select student_fname,department_name,dob from students natural
join departments where student_fname='%s'"%x
    cursor.execute(query)
    result = cursor.fetchall()
    for i in result:
        a=i[0]
        b=i[1]
        c=i[2]
        year=odd_even(c.year)
        print(a+', '+b+', '+year)
    cursor.close()
except(Exception, psycopg2.DatabaseError) as error:
    print(error)
finally:
    connection.close()

```